

Gwangyeol Yu

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Education

Korea University Kookmin University , B.S., AI, Bigdata & Management	Mar. 2026 ~ Mar. 2019 ~ Aug. 2025
• Minor: School of Software • Interdisciplinary Major: Data Science • Overall GPA: 4.04 / 4.5 • Major GPA: 4.09 / 4.5	
Daegwang High School	Mar. 2016 ~ Feb. 2019

Research Interests

Generative modeling | DDPM, Stable Diffusion, 3D Generative
Computer Vision | Object Detection, Tracking, OCR, Segmentation, Video, eXplainable AI
Anomaly Detection | Time Series, Video, Image
Speech | Voice Conversion, Voice Defense

Publications

Mimic Blocker: Self Supervised Adversarial Training for Voice Conversion Defense with Pretrained Feature Extractors
Gwangyeol Yu, Junhyuk Lee, Seoryeong Kim, Jimin Lee, Jehyuk Lee * (* indicates corresponding author)
(scheduled, Interspeech 2025)

Distortion Measurement Metric for Human Image Refinement and Evaluation Using Distorted Image Datasets
Gwangyeol Yu, Kyungseo Yoon, Sangwoo Lee, Yeju Shin, Junghyuk Park * (* indicates corresponding author)
(IEEE ACCESS 2025)

Patents

Development and contentization of athlete fitness indicators using STT and Object Tracking
Gwangyeol Yu, Seoyeon Lee, Kiseoup Shin, Seunghak Lee
KR-Application No. 10-2024-0007425

Professional/Research Experiences

One Shot Voice Conversion Attack (Undergraduate research student at UROP , Kookmin University)	Dec. 2024 ~ Feb. 2025
• Developed a noise generation algorithm that disrupts voice conversion by using style voice inputs, while ensuring the noise remains imperceptible to the human ear.	
Self-Supervised Learning for Video Anomaly Detection (Undergraduate research student at Alpha Project , Kookmin University)	Aug. 2024 ~ Dec. 2024
• Conducted research on video anomaly detection using self-supervised learning with a driver video dataset • Developed an anomaly detection approach leveraging jigsaw-puzzle-based techniques	

Transformer based MTA Model

Mar. 2024 ~ Jun. 2024

(Undergraduate Research Student in the University-Industry Collaboration Research Program, Kookmin University&Nasmeida)

- Proposed a Transformer-based Multi-Touch Attribution model leveraging user data to improve attribution accuracy
- Focused on building a reliable and interpretable model with explainable attribution capabilities

Improving Diffusion Model Performance through Explainable AI

May. 2024 ~

(Research Student in the X:AI(AI Academic Society) Research Session, Kookmin University)

- Applied Explainable AI (XAI) techniques, such as Grad-CAM, to diffusion models to detect decision-critical regions and improve model performance

Developing Technology for Detecting Personal Information in Images

Apr. 2022 ~ Nov. 2022

(Fellow in the SKT AI Fellowship, SK Telecom)

- Developed a model to detect personal information (names, addresses, faces) in documents containing sensitive data
- Focused on generating personal information datasets and detecting key attributes within images

Awards & Honors

HD Hyundai AI Challenge (Grand prize) Hyundai	Dec. 2023
2023 BigContest (Grand prize) Ministry of Science and ICT, NIA	Dec. 2023
Hackathon Competition for the Disabled (Disability Plus Technology, Honorable mention) Ministry of Health and Welfare	Oct. 2023
DACrew 4th (Encouragement Award) DACON	Feb. 2023
KAMP: The 2nd K-Artificial Intelligence Manufacturing Data Analysis Competition in 2022(Gold prize) KAIST, KOSMO, Ministry of SMEs and Startups	Dec. 2022
2022 BigContest (gold prize) Ministry of Science and ICT, NIA	Dec. 2022
SKT AI Fellowship 4th (1st,Encouragement prize) SK Telecom	Nov. 2022
The 2nd NH Investment & Securities Big Data Competition(Honorable mention) NH Investment & Securities Co., Ltd	Nov. 2021
2021 BigContest (Encouragement Award) Ministry of Science and ICT, NIA	Dec. 2021

Projects

Generating Facial Caricature Profile GIFs X:AI	Jul. 2024 ~ Aug. 2024
Accident Risk Prediction (ECLO) Using Spatiotemporal Information Ministry of Trade, Industry and Energy	Nov. 2024 ~ Dec. 2024
Port Ship Waiting Time Prediction Hyundai	Sep. 2024 ~ Nov. 2024
Weight Prediction Using Excavator Sensor Data Hyundai	Sep. 2024 ~ Nov. 2024
Image IQA Assessment and Captioning Samsung Advanced Institute of Technology	Aug. 2023 ~ Oct. 2023
Generating 3D Shoes Customization X:AI	Jul. 2023 ~ Nov. 2023
Path Guidance for Visually Impaired Individuals Using Object Detection and Segmentation Ministry of Health and Welfare	Jul. 2023 ~ Oct. 2023
Emotional Diary: Emotion Analysis and Personalized Recommendations Google Developer Student Clubs, Korea	Jan. 2023 ~ Mar. 2023
Process Anomaly Detection Through Time Series Analysis KAIST, KOSMO, Ministry of SMEs and Startup	Oct. 2022 ~ Dec. 2022

Loan Application Prediction Analysis Using App Usability Data Ministry of Science and ICT, NIA	Sep. 2022 ~ Dec. 2022
Hairstyle Recommendation and Generation System Based on Facial Shape X:AI	Jul. 2022 ~ Dec. 2022
Flood Prediction Using Dam Inflow Forecasting Ministry of Science and ICT, NIA	Sep.2021 ~ Dec. 2021
Stock Holding Period Prediction and Service Suggestions NH Investment & Securities Co., Ltd.	Aug. 2021 ~ Nov. 2021

Additional Experiences

X:AI AI Academic Society, Kookmin University	Jan. 2022 ~ Dec. 2024
1st - 3rd (Vice president): Base, Adv, Poster, Writing session 4th (President) : Base, Adv, Poster, Writing session 5th (Vice president): Base, Adv, Poster, Writing session	
Kaggle Korea study Sharing knowledge and resources a group of AI enthusiasts	Jan. 2022 ~ Dec. 2024
Google Developer Student Clubs Google Developer Groups	Jun. 2022 ~ Mar. 2023

Certifications

SQLD Korea Data Agency	08 Apr. 2022
ADsP Korea Data Agency	29 Sep. 2020

Skills and Techniques

Programming Languages | Python, R, C++ , Java, javaScript

AI & ML Frameworks | Pytorch, TensorFlow, Scikit-learn, Transformers, PyTorch-Lightening, Numpy, Pandas, Matplotlib, Seaborn

Development Tools & Platforms | VSCode, PyCharm, Colab, Jupyter Notebook/Lab, Docker, Wandb, Git, GitHub

Others | Gradio, Streamlit, QGIS, MySQL, Slack